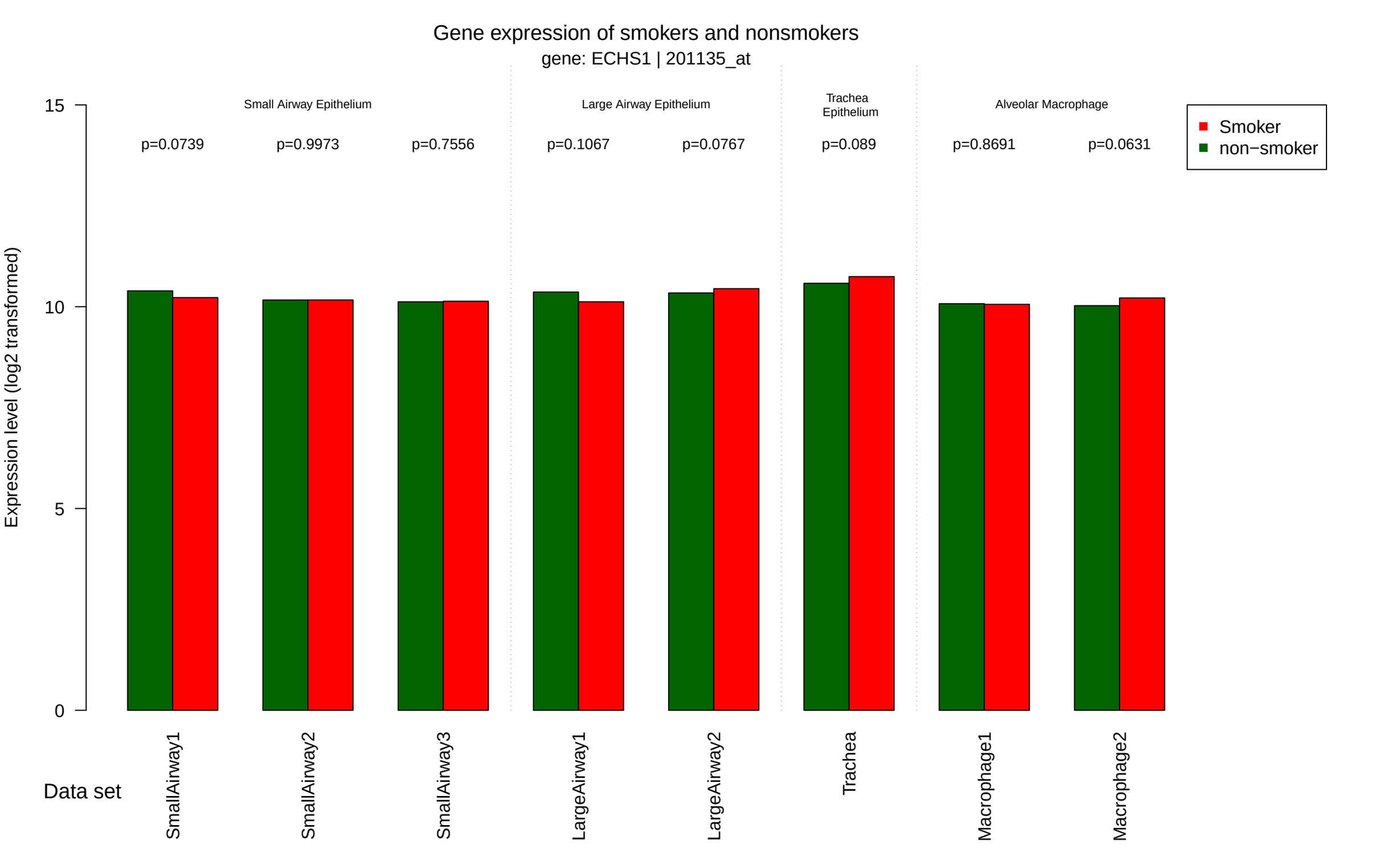
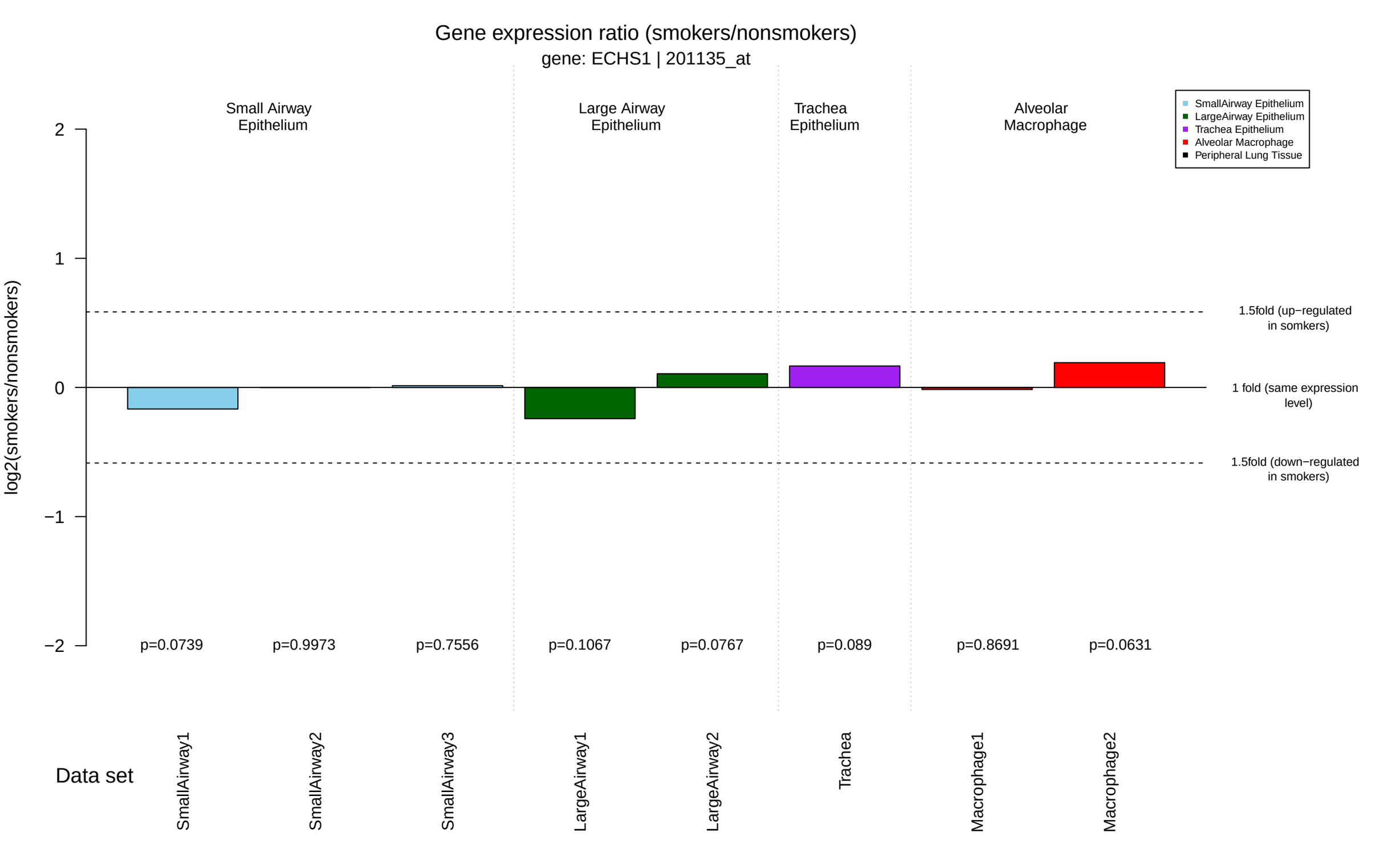
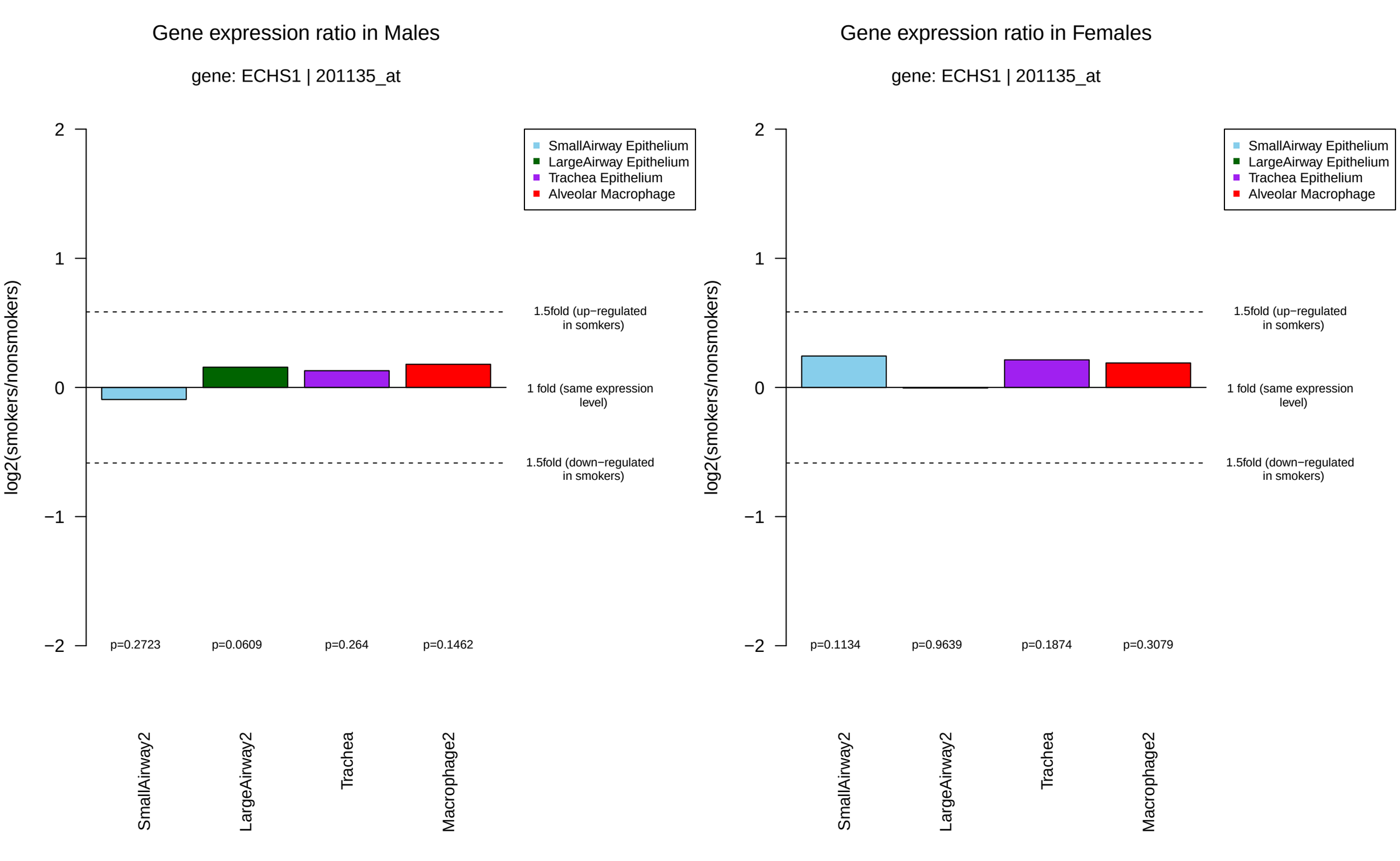
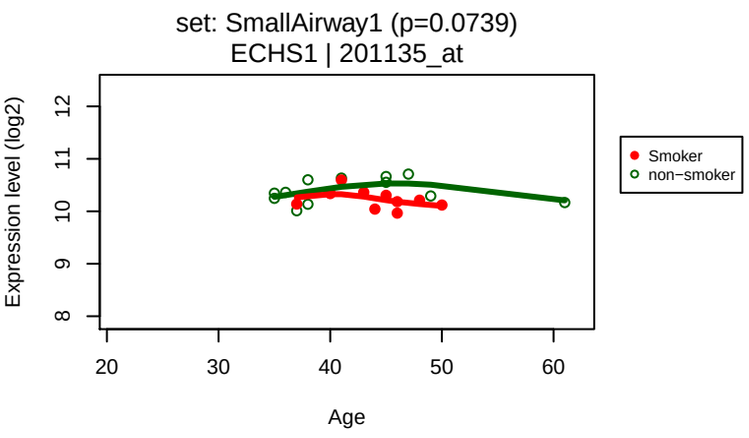
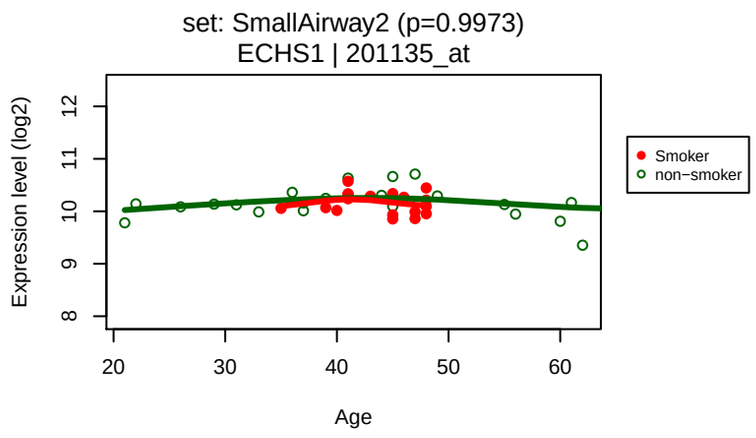
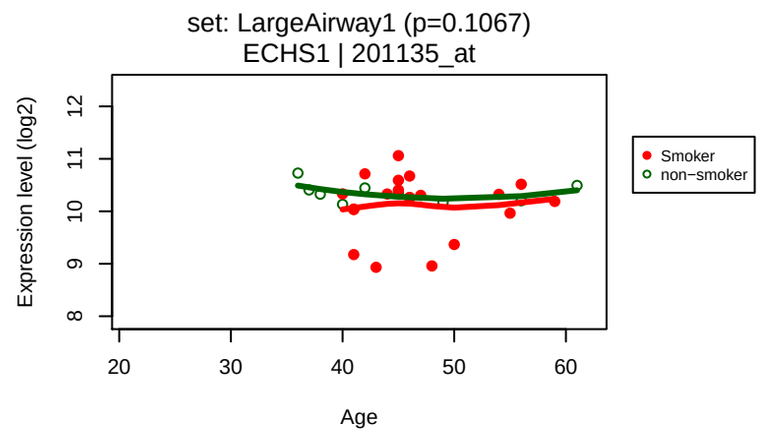

Gene expression of smokers and nonsmokers
gene: ECHS1 | 201135_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0739
p=0.9973
p=0.7556
p=0.1067
p=0.0767
p=0.089
p=0.8691
p=0.0631
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

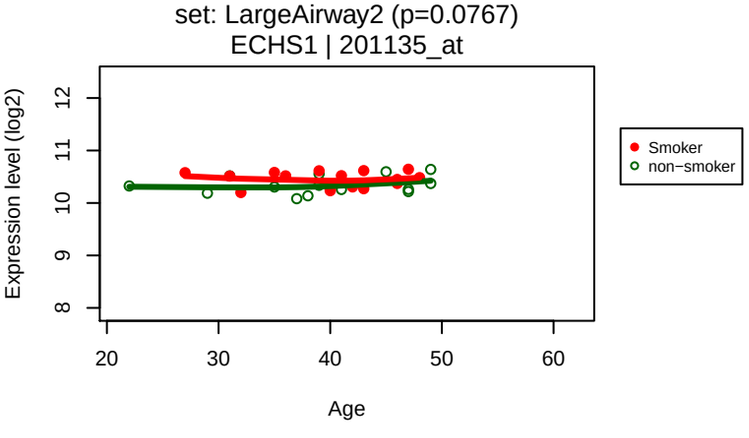

Gene expression ratio (smokers/nonsmokers)
gene: ECHS1 | 201135_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0739
p=0.9973
p=0.7556
p=0.1067
p=0.0767
p=0.089
p=0.8691
p=0.0631
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

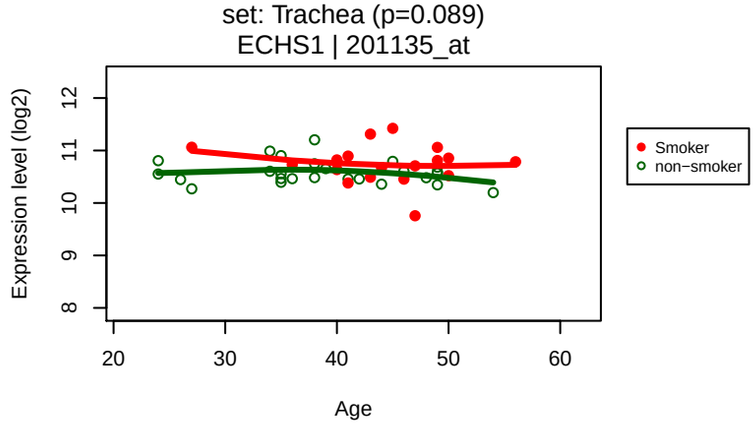

Gene expression ratio in Males
gene: ECHS1 | 201135_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2723
p=0.0609
p=0.264
p=0.1462
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ECHS1 | 201135_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1134
p=0.9639
p=0.1874
p=0.3079
SmallAirway2
LargeAirway2
Trachea
Macrophage2

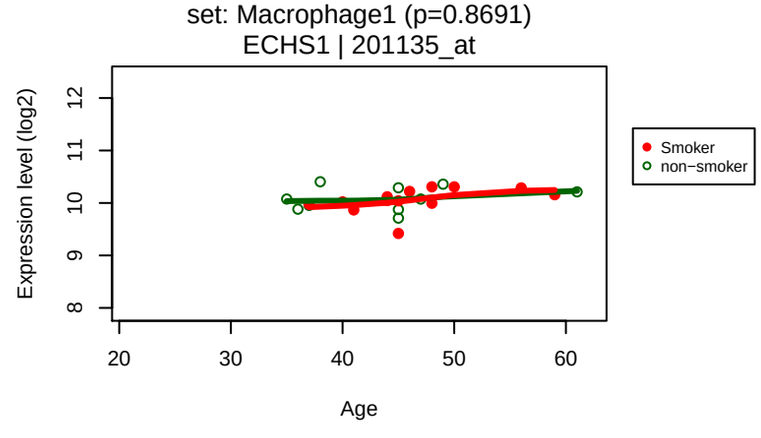

set: SmallAirway1 (p=0.0739)
ECHS1 | 201135_at
Expression level (log2)
Age
Smoker
non−smoker

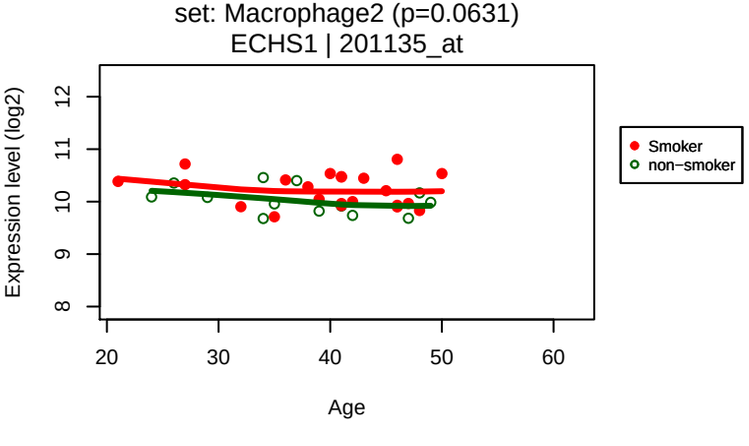

set: SmallAirway2 (p=0.9973)
ECHS1 | 201135_at
Expression level (log2)
Age
Smoker
non−smoker

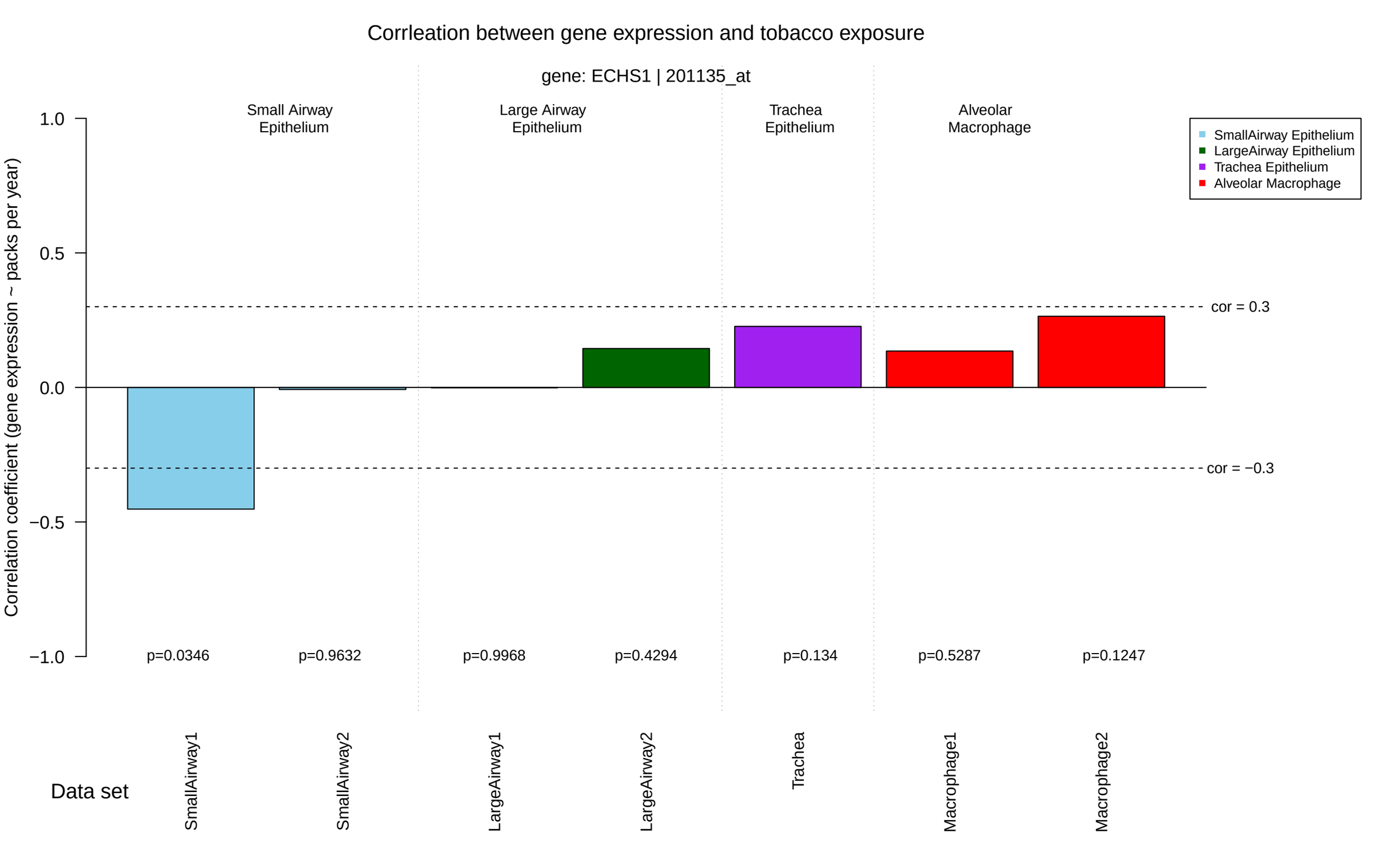

set: LargeAirway1 (p=0.1067)
ECHS1 | 201135_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0767)
ECHS1 | 201135_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.089)
ECHS1 | 201135_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8691)
ECHS1 | 201135_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0631)
ECHS1 | 201135_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ECHS1 | 201135_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0346
p=0.9632
p=0.9968
p=0.4294
p=0.134
p=0.5287
p=0.1247
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2